

A Brief Discussion of Free Merge, Markovian Merge, and Non-Markovian Merge

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Abstract: This paper is principally concerned with the principle of Free Merge, arguably one of the central commitments of the Minimalist Programme. It first traces the relevant theoretical developments, before turning to a more gradual analysis of three notions: Free Merge, Markovian Merge, and Non-Markovian Merge. The discussion will then consider what kinds of problems this theoretical apparatus may be able to address, while also offering an exposition and interpretation of Chomsky's most recent proposals.

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In the early Minimalist Programme, as represented by Chomsky (1993, 1995), and indeed in much earlier Chomskyan formal linguistics, one rather salient assumption was that Merge—whether External Merge or Internal Merge—was **purpose-driven**. External Merge, for instance, was taken to satisfy selectional features, while Internal Merge was commonly understood as motivated by some derivational requirement, such as feature-checking. From Chomsky (2004) onwards, however, External Merge and Internal Merge came to be unified under the single operation **Merge**, and Chomsky began to suggest that this operation was free.¹ With

¹ Chomsky first advances the notion of **Free Merge** within his own theoretical architecture: Narrow Syntax (NS) is based on the free operation Merge. The SMT entails that the merger of α and β is unconstrained, and may therefore be either External Merge or Internal Merge. Under External Merge, α and β are distinct objects; under Internal Merge, one is contained within the other, and Merge yields the

Chomsky (2013, 2015), namely *Problems of Projection* and *Problems of Projection: Extensions*, the hypothesis of **Simplest Merge** was introduced, and the freedom of Merge received a more sustained formulation. Chomsky (2015), for example, assumes that “operations can be free, with the outcome evaluated at the phase level for transfer and interpretation at the interfaces.” Samuel David Epstein, Hisatsugu Kitahara and T. Daniel Seely (2017: 29) make the point still more sharply: “Merge is completely optional. It applies or it doesn’t, and if it does, it applies for no other reason than because ‘it can’.”

With respect to the definition of Free Merge, the formulation offered by Pan Junnan and Du Yuqiao (2024) seems particularly lucid. On their view, the unification of composition and displacement entails that syntactic generation by movement ought to be free and unrestricted. The reason is straightforward enough: External Merge, which is responsible for composition, is evidently free and unconstrained; if Internal Merge is essentially identical to External Merge, differing only in the object to which it applies, then Internal Merge, responsible for displacement, ought not to bear any additional restriction. This view is what they call the **Free Merge Hypothesis**. Such a definition, in effect, develops a line already implicit in Chomsky (2004), while giving it a more explicit theoretical articulation.

As for the **Simplest Merge** hypothesis advanced by Chomsky (2013, 2015), one should also note that Collins (2002) was the first to propose a minimised conception of Merge in which labels are abandoned. This proposal helped to prepare the ground for Chomsky’s later development of Simplest Merge. Chomsky’s more recent position continues this trajectory: “Merge”, when defined in its simplest form, applies

property of “displacement”. Since displacement is ubiquitous in language, any adequate theory must account for it in some fashion. It is difficult to envisage a simpler approach than allowing Internal Merge—that is, a grammatical transformation—as a freely available operation. Accordingly, displacement is not an “imperfection” of language; rather, its absence would itself constitute an imperfection. Chomsky (2004: 110)

freely — “Merge, defined in the simplest form, also applied freely” (Chomsky et al. 2023: 56). Collins (2017: 51–53) lists thirteen properties of Merge. Of particular relevance here are the fifth and sixth: fifth, no label is encoded in the output of Merge; sixth, Merge is untriggered. In order to calculate Merge(X, Y), no feature needs to be checked, whether a subcategorisation feature, an EPP feature, or some comparable formal property.

A related clarification is provided by Epstein, Kitahara and Seely (2022). The **Strong Minimalist Thesis** (SMT), introduced by Chomsky (1993, 1995a, b) and further developed in Chomsky (2000) and subsequent work, takes the computational system of human language to be a “perfect system”, meeting interface conditions in a manner consistent with third-factor principles. This, of course, is not an “assertion”, but rather a hypothesis deemed worthy of investigation on methodological grounds shared by ordinary scientific inquiry. Within the SMT, then, the combinatorial operation of the generative procedure is, by hypothesis, stated in the simplest possible form: what later came to be called **Simplest Merge**, a set-forming operation which takes two elements X and Y and constructs the set {X, Y}.

Epstein, Kitahara and Seely offer a useful summary of this development. A recent move in minimalist theory, associated with Chomsky (2013, 2015), is to define Merge in its simplest form and to allow it to apply freely, provided that it conforms to third-factor principles. These include the **Inclusiveness Condition**, according to which “no new objects are added in the course of computation apart from arrangements of lexical properties” (Chomsky 1995: 228), and the **No Tampering Condition**, according to which “Merge of X and Y leaves the two SOs unchanged” (Chomsky 2008: 138). On this hypothesis, Merge no longer operates merely “in order to” create a configuration in which interface-illegitimate features may be checked. Rather, Merge applies optionally, and derivational crash is therefore possible. Chomsky (2015) accordingly assumes that “operations can be free, with the outcome

evaluated at the phase level for transfer and interpretation at the interfaces” (Epstein, Kitahara and Seely 2022: 18).

What, then, is the relation between **Free Merge** and **Simplest Merge**? I would suggest that Free Merge is, in effect, the presuppositional principle of Simplest Merge. That is to say, Free Merge answers a particularly clear question: Merge need not be **purpose-driven** in any substantive sense. It applies simply because “it can”. Simplest Merge, by contrast, is the operational form which Merge assumes under the principle of Free Merge. The core of Chomsky (2015) and Chomsky et al. (2023) lies precisely in the definition of Simplest Merge as a specific operation; Free Merge, however, names the intrinsic property of Merge itself. It does not require any independent justification for its existence. It is merely the operation that combines two elements into one syntactic object, and it does so freely, optionally, and without teleological motivation. This, I take it, is the force of the phrase “it can”.

In Chomsky (2019), the notion of the **Workspace** is introduced. Chomsky observes that “one immediate consequence is that operations, say, the right version of Merge should be operations on the workspace, not on a particular syntactic object, because they can change the workspace” (Chomsky 2019: 275). Merge is thus relocated within the Workspace.

Chomsky (2019a) introduces the **Workspace** (WS) and defines Merge as an operation applying within it, denoted as **MERGE**. Yet Chomsky (2019b) specifically notes that this MERGE is subject to stricter restrictions than earlier versions of the recursive mechanism. The question, then, is what these restrictions amount to. Chomsky’s treatment in 2019b begins by distinguishing recursion in language from recursion in the more general sense, and he attributes the relevant difference to **Restricting Resources** (RR), an issue taken up again in Chomsky (2023).

Chomsky (2023: 16) writes that Merge is “a computational device, one that is a component of an organic system, and thus is constrained in its application by

restrictions imposed by organic systems and by general principles governing the operation of any computational mechanism; it is, in short, necessarily subject to the third factor.” Here it is worth recalling that the **third factor** contains several subtypes: “(a) principles of data analysis that might be used in language acquisition and other domains; (b) principles of structural architecture and developmental constraints that enter into canalization², organic form, and action over a wide range, including principles of efficient computation, which would be expected to be of particular significance for computational systems such as language” (Chomsky 2005: 6).

What should be stressed is that the third factor is, in substance, a kind of biological constraint: a principle independent of the language faculty, but not thereby irrelevant to it. As Cheng Gong (2022) puts it, the “third factor” refers to the physical and mathematical laws to which all biological systems must conform. What, then, are the third-factor restrictions on Free Merge? I would propose that they operate along two dimensions.

First, there are constraints of **biological reality**. Fong Sandiway (2021) argues that the human brain, as a slow organic system, lacks the requisite memory storage capacity and must therefore selectively discard a substantial amount of signal information.³ Human cognition is consequently constrained by the physical limitations of the brain. Such biological constraints are, in effect, what is meant by **Restricting Resources**.

Second, there are **principles of efficient computation**. These may be taken to include several mechanisms:

- (a) the **Markovian Property**;
- (b) **Minimal Search**;

² In evolutionary genetics, canalisation refers to the tendency of long-term natural selection to produce optimised phenotypes, thereby reducing phenotypic variation. Cheng Gong and Shen Yuan (2022: 34, n. 1)

³ The memory-storage space referred to here is, in psychological terms, **short-term memory**, whose temporary storage capacity is highly limited.

- (c) **Minimal Yield**;
- (d) the **No Tampering Condition** (NTC);
- (e) the **Inclusiveness Condition**.

The **Markovian Property** and **Minimal Search** will be discussed below. For the moment, the remaining mechanisms may be introduced as follows.

Minimal Yield refers to the requirement that each application of Merge produce only the necessary new set, rather than preserving the old inscriptions that have already been merged. Since the number of new inscriptions in the Workspace after Merge should be kept as small as possible, in accordance with third-factor considerations such as the brain's resource limitations, Merge will generally introduce only one new inscription.

The **No Tampering Condition** (NTC) states that Merge may neither modify nor delete the internal structure of a syntactic object already generated. The two elements merged must remain unchanged after the operation.

The **Inclusiveness Condition** holds that the computational procedure may only rearrange the already available features of lexical items. It may not introduce new objects, such as traces, bar-level projections, or comparable theoretical artefacts.

We may now ask whether the foregoing claim—that Merge is subject to restrictions—stands in tension with the thesis of Free Merge. At the same time, Chomsky et al. (2023), especially in Chapter 5, maintain that Merge must not only conform to third-factor principles, but must also obey the **first factor**, namely language-specific conditions (**LSC**), in particular **Theta Theory** (TT). Around this point, the authors further propose the principle of the **Duality of Semantics**, according to which Internal Merge constructs non-argument positions, or A'-positions, whereas External Merge constructs argument structure, or A-positions.⁴ They also

⁴ The **A'-positions** constructed by Internal Merge are chiefly associated with additional layers of information structure or functional interpretation. By contrast, the

argue that the Computational System for Human Language—language in the narrow sense—must satisfy language-specific conditions. Its products, correspondingly, must receive phonological and semantic interpretation at two interfaces: the **Conceptual-Intentional Interface (C-I)** and the **Sensorimotor Interface (SM)**.

At this point, the question becomes still more pressing: do these proposals conflict with Free Merge? My answer would be that they do not. There are two reasons.

First, the constraints imposed by the third factor are biological in character: they concern the physical limitations of the brain, the efficiency of computation,⁵ and the principled architecture of Universal Grammar, including LSC. None of this excludes the claim that Merge is free in its selection of objects. Rather, Merge is constrained to operate within the Workspace. Under Free Merge, syntactic operations may apply to all elements available in the Workspace and may thereby generate any structure that the system is capable of generating.

Second, the structures so generated must ultimately be evaluated and filtered at the interface level. For the interface to recognise a structure, labelling must render it interpretable. Put differently, Merge need not be triggered in order to apply; yet the products of Merge are subject to first-factor constraints when they are submitted to interface interpretation. Free Merge is therefore best understood as **freedom of process**, not unconstrained legitimacy of output. Its outputs remain subject to evaluation.

A-positions constructed by External Merge are generally projected by functional heads and conform to the **Argument Criterion** and the **Theta-Criterion** (Collins 2024). The authors also point out that the **Duality of Semantics** constrains the manner in which Merge may apply.

⁵ The principles of efficient computation seek simplicity, minimisation, and economy. As determinants of computational efficiency, they derive from natural laws of the sort associated with physics and mathematics, and are taken to apply across organic and computational systems alike.

A further point should be made here. Ginsburg (2024) offers a modelling analysis of Free Merge and advances several observations that are, in my view, rather suggestive. Two claims are especially worth endorsing.

First: “I argue that arguments can be subject to Free Merge, subject to the constraints of the language module, but that Labeling in general is sufficient to eliminate most impossible derivations” (Ginsburg 2024: 2).

Second: “I have shown that in general, Labeling, combined with phase boundaries, is sufficient to constrain Free Merge. Also, note that Theta Theory and Case Theory play no clear role in ruling out derivations, and Labeling alone is generally sufficient to constrain Free Merge” (Ginsburg 2024: 53).

These two claims, in effect, reinforce the point that Merge is untriggered and free, while still being constrained by the language module. That module corresponds, in the terms used above, to the first and third factors. Such constraints are not external impositions on the language faculty, but rather properties intrinsic to it. Ginsburg’s emphasis on labelling is, of course, significant. Yet I would maintain that labelling is an algorithm for structural filtering and interface recognition, rather than a device of syntactic construction. The constructive device is one thing only: **Merge**.

On this view, phenomena traditionally associated with first-factor principles, such as **Theta Theory** and **Case Theory**, are better understood as derivative effects of the labelling algorithm and of interface recognition, rather than as independent syntactic modules. In Chomsky’s recent formulation of what he calls the **miracle creed**, he proposes that the element capable of undergoing Internal Merge must be a **θ -marker**. This issue will be taken up below.